Brayden Kushner

412-770-7606 | braydenkushner@gmail.com | linkedin.com/in/braydenkushner

EDUCATION

The Pennsylvania State University — University Park, PA
College of Engineering — May 2027
Bachelor of Science in Computer Engineering — GPA: 3.56
Minor in Cybersecurity Fundamentals — Dean's List: Fall 2023, Spring 2024

WORK EXPERIENCE

Penn State College of Engineering — University Park, PA
Teaching Assistant – Engineering Career Preparation and Development (ENGR 287) — Aug 2026 – Present

- Conducted mock interviews and 1-on-1 resume/LinkedIn coaching sessions for 80+ engineering students
- Graded and provided feedback on career assignments including resumes, cover letters, and LinkedIn profiles
- Supported class instruction with real time input and will serve on an end-of-semester Q&A panel

Bechtel Plant Machinery, Inc. (BPMI) — Monroeville, PA
Electrical Engineering Intern – Embedded and Control Systems for Power Equipment — May 2026 – Aug 2026

- Provided technical reviews on engineering designs, documentation, and system updates to support continuous improvement
- Analyzed inspection and performance data of power systems, and identified trends and inconsistencies to refine reporting accuracy
- Contributed to the development of improved diagnostic software and power electronic testing approaches by evaluating proposals
- Strengthened firmware code and document quality by identifying errors and ensuring alignment with technical requirements

PROJECTS

Data Visualization Application — June 2026 – August 2026
Intern Project at BPMI

- Designed and developed a MATLAB GUI app for rapid analysis of large lab datasets with Fourier and auto-spectrum features
- Interfaced with stakeholders to turn needs into features like variable selection, sample rate controls, and custom output graphs
- Optimized data processing workflows using partial reads, pointer-based data access, and selective plotting for 4x speed increase

Dynamic Storage Allocation Application — March 2026
Operating Systems Design and Construction Coursework

- Partnered with a peer to design and implement a custom C dynamic memory allocator, replacing malloc, free, and realloc
- Evaluated and transitioned between implicit and explicit free list implementations, improving memory utilization by 20%
- Debugged heap corruption and alignment issues using GDB and Valgrind

FPGA Pipelined Processor — May 2025
Computer Organization and Design Coursework

- Engineered a 5-stage pipelined MIPS CPU in Vivado with Verilog, creating modules for ALUs, memory, adders, and control logic
- Integrated forwarding and hazard detection systems to resolve data dependencies and inserted nops to prevent instruction corruption
- Verified correctness by decoding 22 instruction lines, debugging 984 control lines, and analyzing waveforms across 80+ cycles

ACTIVITIES

Co-Director of Mentor Relations, Engineering Orientation Network — June 2025 – Present

- Worked with executive board in biweekly leadership meetings to coordinate program initiatives and mentor engagement strategies
- Oversee team of 70+ mentors, leading recruitment, conducting interviews, and managing onboarding and performance expectations
- Coordinate logistics for freshmen orientation event servicing 2,000+ students, optimizing dismissal time for a ~10% improvement

Volunteer Photographer, Penn State Dance Marathon (THON) Photography Committee — Sept 2024 – May 2026

- Captured quality photos with a focus on storytelling to contribute to THON, which has raised $220 million for childhood cancer
- Employed skills of communication to establish a shot list and timeline for photo editing with event leaders and deliver on time
- Contributed to weekly meetings to discuss upcoming events and to set weekly, monthly, and annual goals

SKILLS & CERTIFICATIONS

Languages: C, Python, Verilog, VHDL, Java
Tools: VS Code, Vim, Git, GDB, Valgrind, MATLAB, Linux, Xilinix Vivado, Beyond Compare, Creo, Wireshark, Microsoft Suite
Concepts: Operating Systems, Memory Management, Data Structure & Algorithms, Object Oriented Programming, Technical Writing
Clearance: Held a Department of Energy L Clearance